Fiscal Year:	FY 2011	**Task Last Updated:**	FY 09/08/2011
PI Name:	Kennedy, Ann R. D.Sc.		
Project Title:	Center of Acute Radiation Research		
Division Name:	Human Research		
Program/Discipline:	NSBRI		
Program/Discipline--Element/Subdiscipline:	NSBRI--Radiation Effects Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Philadelphia	**State:**	PA
Zip Code:	19104-6072	**Congressional District:**	2
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2008 NSBRI-RFA-08-02 Center of Acute Radiation Research (CARR)
Start Date:	09/01/2008	**End Date:**	08/31/2013
No. of Post Docs:	4	**No. of PhD Degrees:**	0
No. of PhD Candidates:	4	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	Note: title change per NSBRI (12/08)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Freund, Gregory (University of Illinois at Urbana-Champaign) Ware, Jeffrey (University of Pennsylvania) Wan, X. (University of Pennsylvania) Weissman, Drew (University of Pennsylvania) Cengel, Keith M.D., Ph.D. (University of Pennsylvania) King, Gregory (Armed Forces Radiobiology Research Institute)		
Grant/Contract No.:	NCC 9-58-RE01801		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	In this research, animal model systems are being utilized to evaluate the space radiation induced acute radiation syndrome (ARS) with vomiting (and/or retching) and white blood cell counts in ferrets, white blood cell counts, fatigue and immune system parameters in mice, and skin injury, with accompanying immune system changes, and white blood cell counts in pigs as the biological endpoints. In addition, the effects of combined exposure to simulated hypogravity and space radiation on blood cell counts and immune functions with respect to both the innate immune system and the acquired immune system in mice are being evaluated in mice. The first few years of the project are focused on determinations of whether certain parts of the ARS occur in animals exposed to space radiation(s) simulating that occurring during a solar particle event (SPE). If problems are observed to exist for the model systems used, later years of the project will be focused on the development of countermeasures for these effects. In these studies, selected drugs will be evaluated in the animal model systems to determine their effectiveness in prevention and/or mitigation of the specific ARS symptoms. The dose response relationships measured for these endpoints will be compared mathematically with published results for human subjects to determine the RBEs and the predictive value of these animal models for bio-dosimetry and for intervention studies to evaluate FDA-approved drugs that can be used for prevention and/or mitigation of the ARS symptoms. Various tissues will be collected from the animals for histopathological examination and for studies by various immunohistochemistry, qPCR and/or immunochemistry techniques to study the underlying mechanisms for the acute radiation syndrome and its prevention and/or treatment by drugs. As part of the studies in year 3 of this grant's activities, countermeasure studies began to identify agents that could prevent and/or mitigate the effects of SPE radiation on ferret emesis and mouse blood cell counts. These investigations have been designed to address Acute Radiation Syndromes - Specific Gaps 1, 2, 3, 5, 6, 7 and 8 identified in NSBRI-RFA-08-02.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	It is well-known that exposure to ionizing radiation at sufficiently high doses can result in various types of adverse biological effects. The countermeasures to be developed as part of this research program have significance for all individuals exposed to high doses of ionizing radiation on earth as well as in space. The countermeasures shown to protect against radiation induced biological effects from space radiation will be equally useful for protection against radiation induced biological effects on earth.
Task Progress:	- Hematology : 1) 70 MeV proton radiation, which leads to a homogeneous dose of radiation in our studies, results in comparable effects on blood cell counts in mice at doses of 25-200 cGy in proton radiation experiments performed at Loma Linda University Medical Center (LLUMC), the University of Pennsylvania (Penn) and the NASA Space Radiation Laboratory (NRSL). 2) In experiments performed with radiation +/- hindlimb unloading (HU) in mice, it was observed that the splenic T lymphocyte population is significantly decreased in the irradiated + HU group (compared to the non-treated control group). The results also indicated that splenic T cells that were isolated and exposed to exogenous activation in the irradiated +/- HU groups had a reduced ability to become activated (compared to the results from the HU group and the non-suspended, sham-irradiated group). - Pig Skin: 1) In collaboration with scientists at Loma Linda University, a beam delivery/modulation system was designed and constructed to produce a solar particle event (SPE)-like dose distribution in pigs that is comparable to 6+12 MeV electron SPE simulations. Animals were irradiated and followed for 60 days following sham, 5 Gy, 7.7 Gy and 10 Gy irradiation (skin doses) with SPE-like protons. Preliminary results demonstrate gross lung and heart damage with pericardial and pleural effusions in animals exposed to 10 Gy, in addition to numerous other adverse effects. - Fatigue: Overall, the fatigue studies (involving the social exploration, submaximal exercise treadmill, locomotor activity and forced swim tests) suggest that the likelihood of SPE-like radiation (administered at both high and low dose-rates) inducing increases in fatigue in astronauts is very low. - Immune System: 1) We observed that a 2 Gy dose of SPE-like radiations results in a transient increase in release of LPS into the circulation, demonstrating potentially pathologic changes in the GI tract. When this amount of radiation is combined with the HU model of spaceflight, an enhanced and prolonged impairment of commensal bacteria containment is observed. The proposed mechanism for this effect is that radiation plus HU leads to breaks in the tight junctions between GI tract epithelial cells, which results in the migration of LPS into the subepithelial tissue. 2) Our preliminary studies have shown that the combination of radiation and HU impairs the generation of an acquired immune response. - Ferret Vomiting/Retching:) Completed the data set for both protons and gamma radiation for high dose rate (HDR) (50 cGy/min.) or low dose-rate (LDR) ((50 cGy/hr.) for the ferret emesis response, 2) Determination of RBE for proton induced emesis compared to gamma induced emesis in ferrets, 3) Preliminary assessment of efficacy of a countermeasure (Zofran, administered orally) for ferret emesis induced by 2 Gy gamma photons and 2 Gy 110 MeV protons administered at high dose-rate.
Bibliography Type:	Description: (Last Updated: 08/24/2020)
Abstracts for Journals and Proceedings	Cengel KA, Wilson JM, Seykora JT, Sanzari JK, Maks C, Yee SS, Ware J, Diffenderfer ES, Weissman D, McDonough JE, Kennedy AR. "The acute toxicity of highly inhomogeneous solar particle event like dose distributions: effects that are not predicted well by extrapolation from homogeneous irradiation data." 18th IAA Humans in Space Symposium, Houston, Texas, April 11-15, 2011. 18th IAA Humans in Space Symposium, Houston, Texas, April 11-15, 2011. , Apr-2011
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Abstracts for Journals and Proceedings	Kennedy AR, Weissman D, King G, Freund G, Wagner E, McDonough J, Miller A, Mick R, Cengel K. "Acute radiation effects resulting from exposure to solar particle event-like radiation." 18th IAA Humans in Space Symposium, Houston, TX, April 11-15, 2011. 18th IAA Humans in Space Symposium, Houston, TX, April 11-15, 2011. , Apr-2011

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